

Notes on Isolation Forest: a geometric point of view

Matteo Marco Costa

September 4, 2026

Abstract

These notes present a simple visualization framework for the Isolation Forest anomaly score function which can be generalized to other models providing a numerical score function.

1 A visualization approach

Anomaly detection is used in applications such as fraud detection, network security, and system monitoring. However, interpreting how an anomaly score behaves across a high-dimensional feature space is not straightforward. These notes present a simple visualization framework for the respective anomaly score function which can also be extended to other models which adopt a score function:

- **feature selection:** let $\mathbf{X} \in \mathbb{R}^{n \times d}$ denote the dataset, where n is the number of observations and d is the number of features. We define a variability measure $V(X_j)$ for each feature X_j , $j = 1, \dots, d$. The two selected features are those corresponding to the two largest values of V :

$$V(X_{j_1}) \geq V(X_{j_2}) \geq V(X_j), \quad \forall j \notin j_1, j_2.$$

For example, $V(X_j)$ may be chosen as the empirical variance, defined by

$$V(X_j) = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2, \quad \bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}.$$

The selected features are denoted by X_{j_1} and X_{j_2} . This selection is used only to define the two-dimensional representation of the observations in the subsequent visualization and is not intended to identify the features that are most important for the Isolation Forest.

- **data preprocessing and model training:** before the fitting, the input data are transformed or standardized according to the adopted preprocessing procedure. Let $\mathbf{X}^* \in \mathbb{R}^{n \times d}$ denote the resulting dataset. The Isolation Forest is then trained on $\mathbf{X}^*$ using the adopted model and hyperparameter configuration.
- **anomaly score:** after training, the fitted Isolation Forest is used to compute an anomaly score for each observation. Let $\mathbf{x}_i^* \in \mathbb{R}^d$ denote the i -th preprocessed observation. Its corresponding score is given by $s_i = s(\mathbf{x}_i^*)$ for $i = 1, \dots, n$ where $s : \mathbb{R}^d \rightarrow \mathbb{R}$ denotes the score function of the trained Isolation Forest. For example, in Python, the `score_samples` function in `scikit-learn` returns scores for which lower values correspond to observations considered more anomalous. For visualization purposes, the sign of the score can be reversed as $a_i = -s_i$. In this way, larger values of a_i indicate observations considered more anomalous by the model.
- **visualization:** the original values of the two selected features, X_{j_1} and X_{j_2} , define the horizontal and vertical coordinates of the observations, respectively, while the anomaly score determines the color assigned to each observation. The scores are computed using the full preprocessed feature vectors $\mathbf{x}_i^*$, so that all features are taken into account by the Isolation Forest. The two-dimensional representation is therefore used only to visualize the observations and their corresponding model scores.

1.1 A numerical example

We consider a synthetic dataset with two dominant features and a small fraction of anomalous observations. Normal observations are generated from a Gaussian distribution and anomalous ones are generated randomly in the feature space. The Isolation Forest is trained on standardized data using `scikit-learn`. We project the data onto the two selected features and evaluate the fitted score function over a regular grid, keeping the remaining features fixed at representative values. The score sign is reversed for visualization so that higher values indicate more anomalous regions. The resulting score landscape is shown using heatmaps and three-dimensional surfaces, allowing us to inspect whether regions that are distant from the main data cloud are associated with higher anomaly scores.

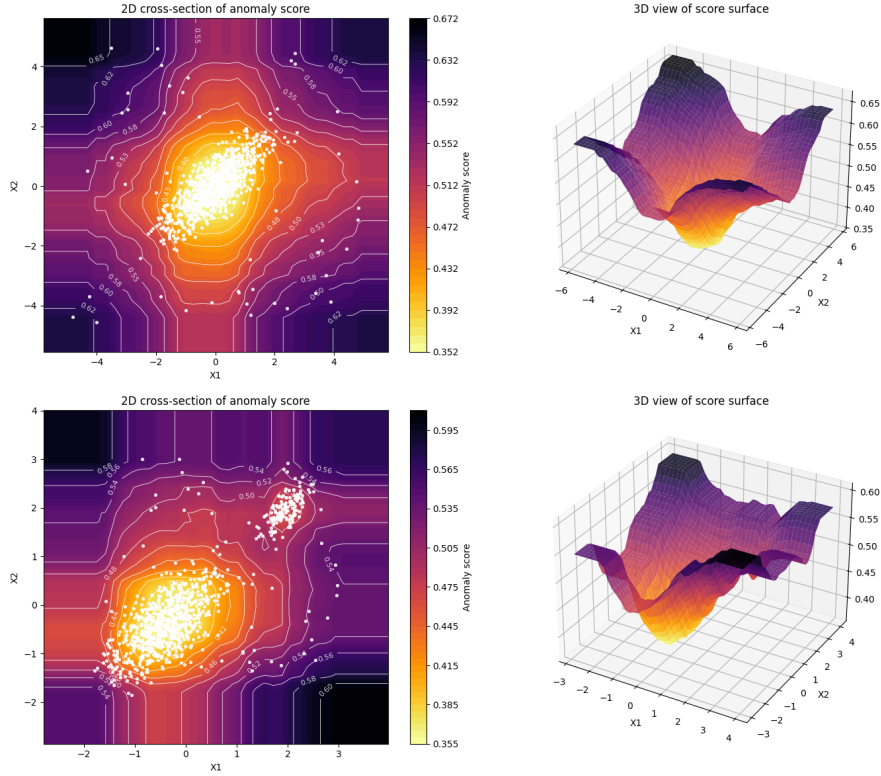


Figure 1: In this example, the visualization suggests that observations located farther from the main data cloud tend to lie in regions associated with higher anomaly scores.

2 Limitations

The proposed approach is intentionally simple and should be regarded as an exploratory visualization rather than a complete interpretability method. Some limitations should be kept in mind:

- **exploratory nature:** the visualization can provide useful intuition and suggest potentially relevant patterns or interactions but these observations should be further investigated using more formal feature-analysis methods;
- **feature selection:** the two features do not necessarily need to be selected according to a specific statistical criterion. They can also be chosen manually based on the particular question or application of interest. For example, one may choose to visualize two features because their relationship is of practical interest;
- **two-dimensional representation:** the visualization only shows the score function with respect to two features at a time. It therefore does not represent the complete score function in the original high-dimensional feature space, and patterns involving other features or higher-order interactions may not be visible.

Conclusion

These notes present a simple way to visualize the behavior of an Isolation Forest. By selecting two features and varying them while keeping the remaining features fixed, the approach provides a two-dimensional view of the anomaly score function. Although it does not capture the full high-dimensional score landscape, the visualization can help reveal patterns or regions of interest that may be difficult to identify from the numerical scores alone. These patterns can then be further investigated using more formal analysis or interpretability methods.